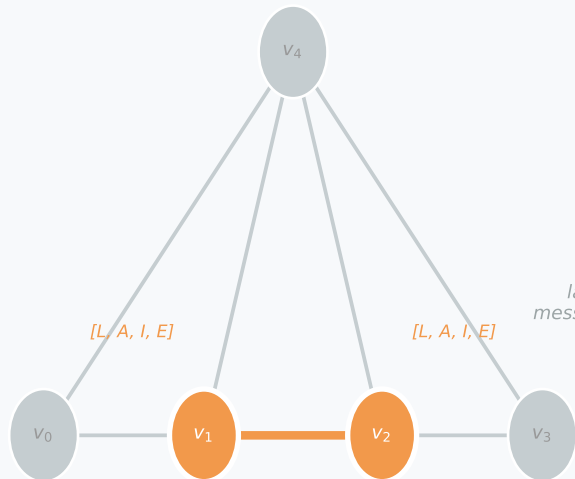


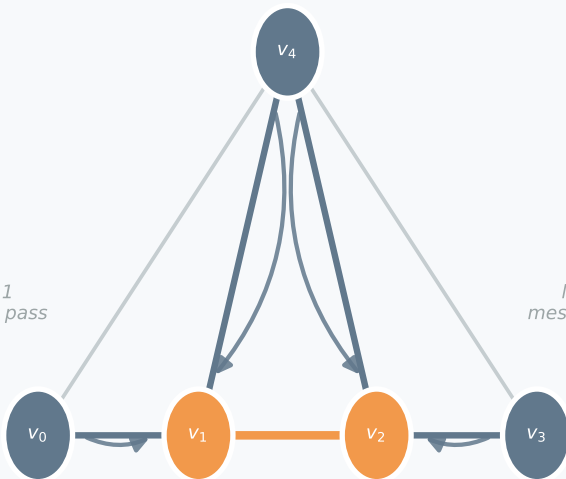
GNN — message passing expands receptive field across truss topology

$k=0$ · initial features



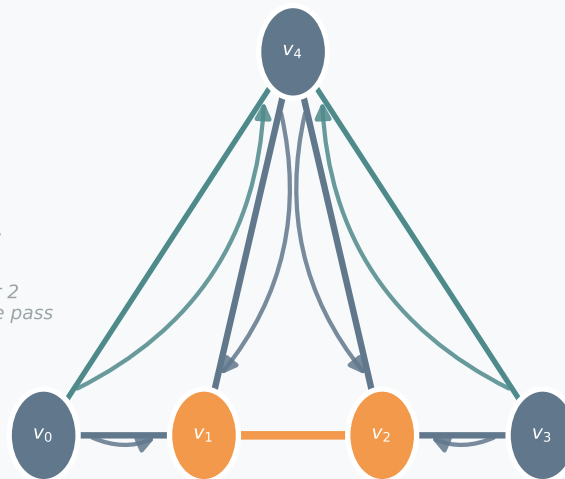
each node holds its own features only

$k=1$ · 1-hop aggregation



target nodes absorb direct neighbours

$k=2$ · 2-hop aggregation



full load-path context reached in two layers

MLP (no context)
 e_{tgt} (isolated)



$[L, \alpha, A, E-I, \dots]$

per-member features only

dense 64

dense 64

output 1

safe / unsafe

no load-path information

neighbour context discarded

target member e_{tgt} 1-hop (direct neighbours) 2-hop (neighbours of neighbours) outside receptive field